9.1 Current & Potential Difference

Question Paper

Course	CIE A Level Physics
Section	9. Electricity
Topic	9.1 Current & Potential Difference
Difficulty	Medium

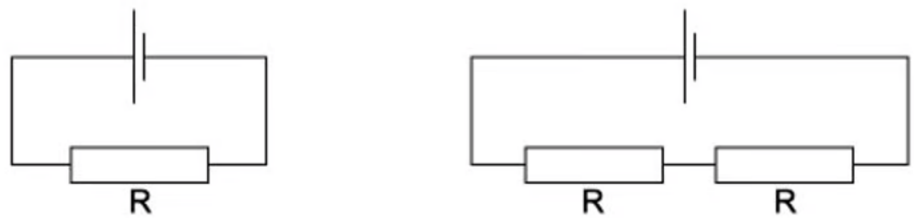
Time allowed: 20

Score: /10

Percentage: /100

Question 1

The diagrams show two different circuits.



The cells in each circuit have the same electromotive force and zero internal resistance. The three resistors each have the same resistance *R*.

In the circuit on the left, the power dissipated in the resistor is *P*.

What is the total power dissipated in the circuit on the right?

A. $\frac{P}{4}$

B. $\frac{P}{2}$

C. P

D. 2P

[1 mark]

Question 2

The potential difference across a component in a circuit is 2.0 V.

How many electrons must flow through this component in order for it to be supplied with 4.8 J of energy?

A. 2.6×10^{18}

B. 1.5×10^{19}

C. 3.0×10^{19}

D. 6.0×10^{19}

[1 mark]

Question 3

Two lamps are connected in series to a 250V power supply. One lamp is rated 240V, 60W and the other is rated 10V, 2.5W.

Which statement most accurately describes what happens ?

A. both lamps light at less than their normal brightness

B. both lamps light normally

C. only the 60 W lamp lights

D. the 10 V lamp blows

[1 mark]

Question 4

Two copper wires of the same length but different diameters carry the same current.

Which statement about the flow of charged particles through the wires is correct?

A. charged particles are provided by the power supply, therefore, the speed at which they travel depends only on the voltage of the supply

B. the charged particles in both wires move with the same average speed because the current in both wires is the same

C. the charged particles move faster through the wire with the larger diameter because there is a greater volume through which to flow

D. the charged particles move faster through the wire with the smaller diameter because it has a larger potential difference applied to it

[1 mark]

Question 5

A power cable X has resistance R and carries current I.

A second cable Y has resistance 2R and carries current $\frac{I}{2}$

What is the ratio $\frac{power\ dissipated\ in\ Y}{power\ dissipated\ in\ X}$?

A. $\frac{1}{4}$

B. $\frac{1}{2}$

C. 2

D. 4

[1 mark]

Question 6

A milliammeter shows a reading of 20 mA.

How many electrons flow through the milliammeter in 10 seconds?

A. 0.20

B. 3.2×10^{20}

C. 200

D. 1.3×10^{18}

[1 mark]

Question 7

In a simple electrical circuit, the current in a resistor is measured as (2.50 ± 0.05) mA. The resistor is marked as having a value of 4.7Ω ± 2 %.

If these values were used to calculate the power dissipated in the resistor, what would be the percentage uncertainty in the value obtained?

A. 2 %

B. 4 %

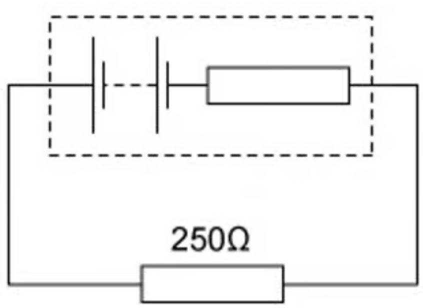
C. 6 %

D. 8 %

[1 mark]

Question 8

A battery, with a constant internal resistance, is connected to a resistor of resistance 250 Ω, as shown.



The current in the resistor is 40 mA for a time of 60 s. During this time 6.0 J of energy is lost in the internal resistance.

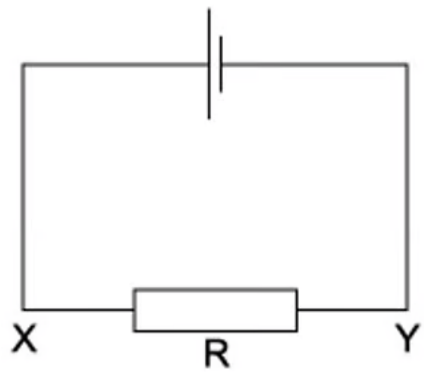
What is the energy supplied to the external resistor during the 60 s and the e.m.f of the battery ?

	Energy / J	E. m. f / V
A	2.4	2.4
B	2.4	7.5
C	24	10.0
D	24	12.5

[1 mark]

Question 9

The current in the circuit shown is 4.8 A



What is the direction of flow and the rate of flow of electrons through the resistor *R* ?

	Direction of flow	Rate of flow
A	X to Y	$3.0 \times 10^{19}\ s^{-1}$
B	X to Y	$6.0 \times 10^{18}\ s^{-1}$
C	Y to X	$3.0 \times 10^{19}\ s^{-1}$
D	Y to X	$6.0 \times 10^{18}\ s^{-1}$

[1 mark]

Question 10

The charge that an electric battery can deliver is specified in ampere-hours.

For example, a battery of capacity 40 ampere-hours could supply, when fully charged, 0.2 A for 200 hours.

What is the maximum energy that a fully charged 12 V, 40 ampere-hour battery could supply ?

A. 1.7 kJ

B. 29 kJ

C. 1.7 MJ

D. 29 MJ

[1 mark]

Model Answers: Medium

1

The correct answer is **B** because:

- Power is equal to $P = VI$
- Current is equal to $I = \frac{V}{R}$
- Combining the two equations gives
  - $P = V\left(\frac{V}{R}\right) = \frac{V^2}{R}$
- so, power dissipated in the circuit on the right is $P = \frac{V^2}{R}$
- The resistance in the circuit on the left is 2R
- So, power dissipated in the circuit on the right is $\frac{V^2}{2R} = \frac{1}{2}\left(\frac{V^2}{R}\right) = \frac{P}{2}$
- Therefore, the power dissipated by the circuit on the right is $\frac{P}{2}$

2

The correct answer is **B** because:

- Potential difference is defined as the amount of electrical energy transferred between two points per unit of charge
  - In equation form this can be expressed as $V = \frac{E}{Q}$
  - Rearranging for charge Q gives $Q = \frac{E}{V}$
  - So, charge is equal to $Q = \frac{4.8}{2.0} = 2.4\ C$
- The total charge is equal to $Q = ne$
  - Where n is the number of electrons and e is the charge of one electron
- So, the number of electrons is equal to $n = \frac{Q}{e} = \frac{24}{1.6 \times 10^{-19}} = 1.5 \times 10^{19}$

3

The correct answer is **B** because:

- One lamp is rated 240 V, 60 W, this means that for the lamp to operate normally, there should be a potential difference (p.d.) of 240 V across it
- Electrical power is equal to $P = VI$
  - So, the current flowing through this lamp $= \frac{P}{V} = \frac{60}{240} = 0.25\ A$
- The other lamp is rated 10 V, 2.5 W. This lamp operates normally when there is a p.d. of 10 V across it
  - Current flowing through the lamp $= \frac{P}{V} = \frac{2.5}{10} = 0.25\ A$
- So, for normal operation, the same current of 0.25 A is required for both lamps. Since the lamps are connected in series, the same current would flow
  - So, both lamps light at their normal brightness
- In the same way, the sum of p.d. across the 2 lamps is equal to 240 + 10 = 250 V

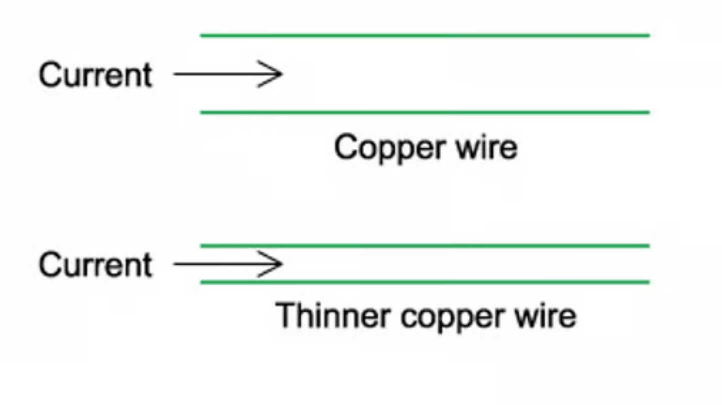
4

The correct answer is **D** because:

- The equation for current in a current-carrying conductor is given by
  - $I = nAve$
  - Rearranging this for *v* we find $v = \frac{1}{nAe}$
- Since the current *I*, number of electrons *n* and charge *e* are the same for both wires, the electrons are faster in the thinner (smaller *A*) wire
  - In the next topic we will look more at resistivity in a wire, for now let's just consider the equation $R = \frac{\rho L}{A}$
- Since length *L* and resistivity *ρ* are the same for both wires, the thicker (larger *A*) wire will have a smaller resistance, so, the thinner wire will have a larger resistance
- Current is equal to $I = \frac{V}{R}$ for each wire, and since the wires have the same current *I*, the value of $\frac{V}{R}$ will be the same in each wire
- This means the thinner wire will have a larger resistance *R* and a smaller potential

difference V across it, while the thicker wire will have a smaller resistance R and a larger potential difference V across it

- Therefore, the charged particles move faster through the thinner wire because it has a larger potential difference applied to it



A is incorrect because charged particles are not provided by the power supply, the charged particles (electrons) are already present in the copper wire

B is incorrect because the equation $\frac{1}{nAe}$ tells us the speed of the electrons depends on the thickness of the wire, so the speed will not be the same

C is incorrect because the equation $\frac{1}{nAe}$ tells us the electrons will be faster in the thinner wire (smaller A, larger v), not the thicker wire (larger A, smaller v)

5

The correct answer is **B** because:

- Power is equal to $P = VI$
- Potential difference is equal to $V = IR$
- Combining the two equations gives
  - $P = (IR)I = I^2R$
- The power dissipated in X is $= I^2R$
- The power dissipated in Y is $= \left(\frac{1}{2}\right)^2(2R) = \frac{2I^2R}{4} = \frac{I^2R}{2}$
  - Therefore, the ratio $\frac{power\ dissipated\ in\ Y}{power\ dissipated\ in\ X} = \frac{I^2R}{2} \times \frac{1}{I^2R} = \frac{1}{2}$

6

The correct answer is **D** because:

- Charge is defined by $Q = I\,t$
- Current $I = 20 \times 10^{-3} A$
- Time $t = 10\,s$
- So, charge $Q = (20 \times 10^{-3}) \times 10 = 0.2\ C$
- The total charge is equal to $Q = ne$
  - Where *n* is the number of electrons and *e* is the charge of one electron
- So, the number of electrons is equal to $n = \frac{Q}{e} = \frac{0.2}{1.6 \times 10^{-19}} = 1.25 \times 10^{18}$
  - Therefore, the number of electrons is 1.3×10^{18}

7

The correct answer is **C** because:

- Power is equal to $P = VI$
- Potential difference is equal to $V = IR$
- Combining the two equations gives
  - $P = (IR)I = I^2R$
- First, we must convert the uncertainty in current to a percentage uncertainty so we can add the percentage uncertainties together
  - Current $I = 2.5A$
  - Uncertainty in current $\Delta I = 0.05\ A$
- So, the percentage uncertainty in current $= \frac{\Delta I}{I} \times 100\ \% = \frac{0.05}{2.5} \times 100 = 2\ \%$
- Since the power dissipated is given by $P = I^2\,R$, the percentage uncertainty in power is $= (2 \times \%\ uncertainity\ in\ I) + (\%\ uncertainity\ in\ R)$
  - Therefore, $\%\ uncertainity\ in\ P = (2 \times 2\ \%) + 2\% = 4 + 2 = 6\ \%$

8

The correct answer is **D** because:

- Power is equal to $P = VI = (IR)I = I^2R$
- The energy transferred in external resistor is equal to $E = Pt = I^2\,R\,t$

- Current $I = 20 \times 10^{-3} A$
- Time $t = 10\, s$
- So, charge $Q = (20 \times 10^{-3}) \times 10 = 0.2\ C$
- The total charge is equal to $Q = ne$
  - Where n is the number of electrons and e is the charge of one electron
- So, the number of electrons is equal to $n = \frac{Q}{e} = \frac{0.2}{1.6 \times 10^{-19}} = 1.25 \times 10^{18}$
  - Therefore, the number of electrons is 1.3×10^{18}

7

The correct answer is **C** because:

- Power is equal to $P = VI$
- Potential difference is equal to $V = IR$
- Combining the two equations gives
  - $P = (IR)I = I^2R$
- First, we must convert the uncertainty in current to a percentage uncertainty so we can add the percentage uncertainties together
  - Current $I = 2.5A$
  - Uncertainty in current $\Delta I = 0.05\ A$
- So, the percentage uncertainty in current $= \frac{\Delta I}{I} \times 100\ \% = \frac{0.05}{2.5} \times 100 = 2\ \%$
- Since the power dissipated is given by $P = I^2\ R$, the percentage uncertainty in power is $= (2 \times \%\ uncertainity\ in\ I) + (\%\ uncertainity\ in\ R)$
  - Therefore, $\%\ uncertainity\ in\ P = (2 \times 2\ \%) + 2\% = 4 + 2 = 6\ \%$

8

The correct answer is **D** because:

- Power is equal to $P = VI = (IR)I = I^2R$
- The energy transferred in external resistor is equal to $E = Pt = I^2\ R\ t$
  - Current $I = 40\ mA = 40 \times 10^{-3}\ A$
  - Resistance $R = 250\ \Omega$
  - Time $t = 60\ s$
- Energy transferred to the resistor $E = (40 \times 10^{-3})^2 \times 250 \times 60 = 24\ J$
  - Therefore, since 6 J of energy is lost in internal resistance, the battery must

supply a total of 6 + 24 = 30 J of energy to the whole circuit

- The energy supplied by battery is $E = QV$
- Where Q is the charge and V is the e. m .f of the battery.
- Charge transferred, $Q = It = (40 \times 10^{-3}) \times 60 = 2.4\ C$
  - So, e. m. f of battery, $V = \frac{E}{Q} = \frac{30}{2.4} = 12.5\ V$

9

The correct answer is **C** because:

- Electrons flow from the negative terminal of the battery, since they are negatively-charged
- The negative terminal of the battery is represented by the shorter line at the symbol of the battery
  - So, the direction of flow of the electrons is from Y to X
- In terms of current, charge is defined as $Q = It$
- In terms of number of electrons, charge is equal to $Q = ne$
  - Where n and e are the number and charge of electrons respectively
- So, $ne = It$ and the rate of flow of electrons is equal to $\frac{n}{t} = \frac{1}{e}$
  - Rate of flow of electrons $= \frac{n}{t} = \frac{I}{e} = \frac{4.8}{1.6 \times 10^{-19}} = 3.0 \times 10^{19}\ s^{-1}$

10

The correct answer is **C** because:

- Electrical power is equal to $P = VI$
- Power is also equal to $P = \frac{E}{t}$
  - Combining the two equations, $P = \frac{E}{t} = VI$
  - So, energy is equal to $E = VIt$
- The battery is described as '12 V, 40 ampere-hour', this means that the e. m .f. of the battery is $V = 12$ V
- From the example given, 40 ampere-hours means that a current of 0.2 A can be supplied for 200 hours or 0.2 A × 200 h = 40 $A\ h$